Structural Non-Isolability in Knowledge-Based Robot Fault Diagnosis

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Abstract: On the voraus-AD benchmark, a diagnosis layer scores 14 failure modes of a hand-written FMEA knowledge base against per-signal deviations. In 514 of 755 anomalous cycles the top score is tied within one of four mode groups that declare the same joints and signal families. Top-1 accuracy ranges from 0.119 to 0.200 over tie-break orders; a prevalence prior scores 0.187. We report the non-isolable group and search for statistics separating it. For the load-torque group they raise held-out within-group accuracy from 0.562 to 0.840. Under a random split over cycles, the search adopts collision statistics that leave-one-setting-out validation does not.

Keywords: Fault isolation and root cause analysis, Structural analysis methods, Diagnosability analysis, Knowledge-based diagnosis, Anomaly detection, Robotics.

1. INTRODUCTION

An anomaly detector on robot telemetry can report that a cycle is abnormal, but not why. A common way to name likely causes is a line-specific knowledge base built from a failure mode and effects analysis (FMEA): each failure mode declares the joints it affects and the signal families in which it shows, a retrieval function scores the modes against the per-signal deviations of the cycle, and the maintenance technician receives a ranked list of candidate causes.

This paper measures which failure modes such a list can isolate on the public voraus-AD benchmark (Brockmann et al., 2024). The declarations of the knowledge base form a fault signature matrix in the sense of Blanke et al. (2016). In this matrix nine of the 14 failure modes form four groups with the same declared joints and signal families: in three groups the columns are identical, and in the fourth, one column contains the other two. The scoring function is continuous, but in 514 of 755 anomalies two or three modes of one group share the highest score, and the reported answer then depends only on the order in which the modes are listed (Sec. 5.2).

Two further findings are about the evaluation. The first concerns the reference for judging accuracy. During development we compared top-1 accuracy with uniform chance. When fault frequencies are skewed, the appropriate reference is a prevalence prior, which ranks the modes and groups by their frequency in the training data and gives this ranking for every cycle. On voraus-AD the prior is better than the retrieval on top-3, top-5, mean reciprocal rank (MRR), and group accuracy under every tie-break order (Sec. 5).

The second concerns validation. The `setting` field of voraus-AD identifies a physical fault instantiation, for example one added weight or one placement of a collision object, and each instantiation is repeated for 4 to 20 cycles. A random split over cycles puts cycles of the same instantiation into both training and test data. For each non-isolable group we search for a *repair*, a statistic of the existing signals that separates the members. The type of split changes the reported accuracy and also which repair is adopted (Sec. 7). Leakage of this kind is known in machine-learning-based science (Kapoor and Narayanan, 2023) and in fault-diagnosis benchmarks (Hendriks et al., 2022); here it occurs in the inner validation that selects the repair. We test both findings on voraus-AD only.

Contributions. (i) The knowledge base analysed as a fault signature matrix, with non-isolable groups reported instead of broken by a tie-break rule. (ii) Two references for a ranked candidate list on this benchmark: a prevalence prior and the exact range of top-1 accuracy over tie-break orders. (iii) A search over predefined statistics of the existing signals for one that separates a group, reported as a certificate with held-out evidence, controls, and a cost class. (iv) Evidence that grouping the validation by fault instantiation changes which repair is adopted. (v) The number of labels from which a flat classifier is more accurate than retrieval followed by a within-group classifier.

2. FAULT SIGNATURES AND ISOLABILITY

Structural diagnosability analysis determines, before any data are seen, which faults a diagnosis system can at best separate, using the structure of the model equations (Blanke et al., 2016; Krysander et al., 2008). The standard object is the fault signature matrix, whose rows are tests or residuals and whose columns are fault modes. Under the exoneration assumption two modes are non-isolable exactly when their columns coincide. Without it isolability is directional: a mode whose column is contained in another mode's column cannot be isolated from that mode (Krysander and Frisk, 2008; Cordier et al., 2004).

The analysis has been applied to public process benchmarks (Düstegör et al., 2006), extended to graded distinguishability under noise (Jung et al., 2015), and related to spectrum-based fault localisation in software (Travé-Massuyès and Wotawa, 2024).

When the signatures do not separate the faults, both signature matching in fault detection and isolation (FDI) and consistency-based diagnosis in the artificial-intelligence (DX) tradition (Reiter, 1987; de Kleer and Williams, 1987) return a set of candidates; for a complete set of tests and without exoneration the two sets coincide (Cordier et al., 2004; Travé-Massuyès, 2014). A ranked list that breaks ties does not report this set. We report instead the non-isolable group of the top-ranked mode, which contains every mode tied with it.

The usual remedy for a non-isolable group is sensor placement: find an additional measurement that makes the group isolable, at the lowest cost (Krysander and Frisk, 2008; Frisk et al., 2009). Here there is no analytical model of the cell, and the rows of the signature matrix are declarations in an FMEA knowledge base (affected joints, signal families, cycle phase), not residuals derived from a model. What we can search are statistics of the existing signals that the deployed representation does not use. This is test selection with data-driven evaluation, close to the hybrid scheme of Jung et al. (2018). Only a statistic that needs a new measurement corresponds to sensor placement, and the evidence for a statistic is held-out accuracy on unseen fault instantiations, not a structural guarantee.

3. SYSTEM AND DATA

Benchmark. voraus-AD (Brockmann et al., 2024) contains machine data from a six-axis robot on a pick-and-place task, with 12 physically induced anomaly categories. We use the 100 Hz release: 2122 cycles, of which 755 are anomalous and 1367 normal. The official training data are 948 normal cycles; the remaining 419 normal cycles are held out. The anomalous cycles fall under 72 `setting` values, each identifying one physical fault instantiation repeated for 4 to 20 cycles (median 10). Three categories have a single instantiation.

Evidence record. Diagnosis is evaluated on all 755 labelled anomalous cycles, that is, assuming perfect detection; in deployment an anomaly detector such as that of Brockmann et al. (2024) would select the cycles. Each of 26 channels (motor torque, torque sensor, joint position, and joint velocity on each of six joints, plus robot and system current) is resampled to 500 points over task progress and standardised against the 948 normal training cycles, $z = (x - \mu)/\max(\sigma, 0.05 \operatorname{range} \mu)$. For the retrieval, the record of a cycle keeps the five channels with the largest peak $|z|$, the dominant joint (largest peak torque deviation in N m), and the action phase at that peak.

Knowledge base and retrieval. The knowledge base is an FMEA encoded as a graph. Liao et al. (2025) automate the construction of such graphs for robots; ours was handwritten by the authors from the voraus-AD anomaly descriptions and has not been reviewed independently. It declares 14 failure modes: 12 map one to one to the benchmark's categories, and backlash and over-temperature have no instances. Each mode declares its joint scope, its signal families, a cycle-phase hint, the direction and time course of its torque deviation, and a recommended check; the deployed score uses the first three. A mode m is scored against a record by $s = J + F + 0.2E + 0.2P + 0.15C$, where J is 1 if the dominant joint lies in the scope of m (0.5 for scope 'any'), F is the Jaccard overlap of signal families, E is the mean over the record's five channels of $\log(14/n)\log(1 + |z|)$, taken as 0 outside the families and joints of m, with n the number of modes that declare that family and joint, P is 1 if the record's phase agrees with the phase hint of m, and C is the fraction of the families of m that E counts. The scoring form and weights were settled during development on these 755 cycles. Choosing the three weights again on each training half of the protocol below gives a different choice on every split and no gain on the test halves in MRR (0.362 against 0.366) or top-5 (0.600 against 0.651).

Protocol. Unless stated otherwise, results are means over five leave-setting-out (grouped) splits: each `setting` is assigned wholly to the training half or to the test half, stratified by category, so a learned component is always tested on instantiations it has not seen. Intervals are t-intervals over the five splits ($t_{0.975,4} = 2.776$). The splits re-partition the same 72 instantiations, so the intervals describe variation between splits on these data, not uncertainty about unseen instantiations. Comparisons between methods are made per split.

4. NON-ISOLABLE GROUPS

The fault signature matrix we analyse has one row per declaration the score reads: joint scope, signal family, and phase. Fig. 1 shows the columns of the nine modes that share joint scope and signal families with another mode. They form four groups: three load-torque modes (axis weight, lost can, varying weight), two distal collisions (foam, carton), two gripping faults (misgrip, invalid grip), and a commutation/over-temperature pair. Every term of the score except phase agreement depends only on scope, families, and the record. The score penalises extra families and the scope 'any', which corresponds to the exoneration assumption for these rows, but it gives the phase hint 'any' credit in every phase: grip (pick or place) and transfer. In the collision, gripping, and commutation groups the phase hints are also equal, so the columns are identical and the members always receive the same score. In the load-torque group the hints differ (axis weight: any phase; lost can: transfer; varying weight: grip), so the column of axis weight contains the other two. Neither payload mode can then be isolated from axis weight, and because a record has one phase, axis weight ties with one or both of them, so no member of the group ever ranks first alone.

In 514 of the 755 anomalies (68.1%) the top score is shared by two or three modes, and every tie lies inside one group. Of these ties, 498 are in the load-torque group: axis weight with lost can in 238 cycles, axis weight with varying weight in 199, and all three in 61 cycles whose peak lies outside the pick, place, and transfer phases. The other 16 are commutation-pair ties. The count is 514 at each of seven phase weights between 0 and 3; the weight changes only which members tie. The collision and gripping groups

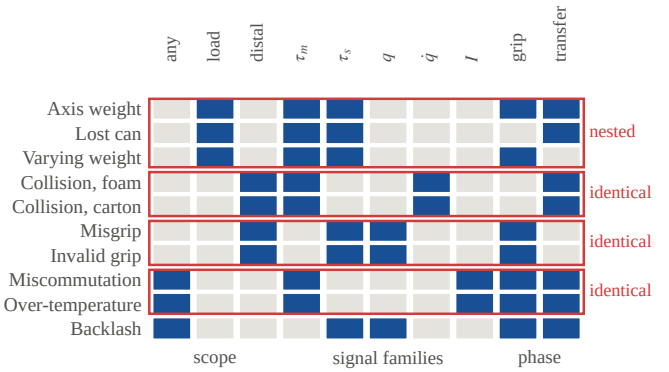


Fig. 1. Signature-matrix columns, drawn as rows, of the nine modes in non-isolable groups and of backlash, which has the gripping pair's families but not their scope. Scope 'load' is J1–J3, 'distal' J4–J6. Families: motor torque τ_m, torque sensor τ_s, joint position q, joint velocity $\dot{q}$, current I; a phase hint 'any' fills both phase cells.

never reach rank one, because they are declared on the distal joints J4–J6, while the dominant joint, chosen by absolute torque deviation, is J1, J2, or J3 in every cycle.

These counts depend on how the knowledge base is written. Declaring the phase of axis weight as transfer instead of any would leave 315 ties. The declared direction and time course of the torque deviation, which the score does not use, differ within every group except the collision pair. Adding them to the score with weight 0.2 removes 513 of the 514 ties, but lowers top-1 on all five splits and balanced recall (Sec. 5.1) on four, from 0.105 to 0.100.

5. TWO REFERENCES FOR THE RANKED LIST

5.1 The prevalence prior

Against uniform chance, 0.071 over 14 modes, the deployed retrieval has a top-1 lift of 2.37×. Uniform chance is the wrong reference when categories differ in frequency: axis weight contributes 156 cycles and cable entanglement 10. The prevalence prior ranks the modes and groups by their frequency in the training half and gives this ranking for every cycle.

Table 1 compares the prior with the retrieval. On top-3, top-5, MRR, and group accuracy the prior is ahead under every tie-break order of Sec. 5.2; for top-3, MRR, and group accuracy it is ahead on all five splits. Top-1 depends on the order. Under the declared order the retrieval is lower by 0.018 and is behind on four of five splits; two of the 12 orders put it at or above the prior. Breaking ties by training-half frequency, which uses only the information the prior uses, gives 0.200, above the prior's 0.187. Group accuracy is also lower for the retrieval: always naming the most frequent group of the training half, which is load torque on every split, scores 0.401 against 0.342.

Balanced recall, the mean of the per-category recalls, is $1/K$ for a constant predictor, whatever the class frequencies; this averages 0.095. The retrieval's balanced recall also depends on the tie-break, so we let tied modes share the credit. The retrieval then reaches 0.105, above the constant on every split but by only 0.002 to 0.016, and one order puts it below. It adds little per-cycle information: its expected recall is zero on six of the 12 categories. Comparisons between methods below use balanced recall in this form.

Table 1. Retrieval and references on the five test halves. Group: accuracy of the predicted mode's group; for the prior, the most frequent group of the training half. Balanced: mean per-category recall, with tied modes sharing the credit; a constant scores $1/K$, with K the 10 or 11 categories in a test half. In parentheses: top-1 with tied modes sharing the credit. All orders: the retrieval's range over the 12 tie-break orders of Sec. 5.2. Two-stage: fixed {sign, step}, Sec. 6; flat classifier: Sec. 8.

	top-1	top-5	group	balanced
uniform chance	0.071	0.357	0.150	0.071
prevalence prior	0.187	0.705	0.401	0.095
retrieval	0.169 (0.160)	0.651	0.342	0.105
all orders	0.119–0.200	0.640–0.651	0.342	0.092–0.119
two-stage	0.309 (0.302)	—	0.342	0.208
flat classifier	0.525	—	0.574	0.397

5.2 The tie-break interval

The scores do not determine which member of a tied set is reported; our implementation reports the member declared first. Only five modes ever tie, always within their group, so only $3! \cdot 2! = 12$ relative orders of the modes can change the result. Table 1 gives the range over all of them: top-1 accuracy runs from 0.119 to 0.200, a width of 0.082. The declared order gives 0.169, and the mean over the orders, 0.160, equals the top-1 accuracy obtained when tied modes share the credit (McSherry and Najork, 2008).

This spread is larger than the one due to the choice of split: over the orders the five-split mean has standard deviation 0.027 and range 0.082, while the declared-order value has standard deviation 0.015 and range 0.040 over the five splits. More splits would not reduce the first, because one order applies to all splits. Both ranges exceed the 0.018 by which the retrieval is below the prior. An oracle that always picks the correct member of the tied set reaches 0.281. This bounds rules that choose inside the tied set, but not rules that choose inside the whole group, because the phase term has already removed the true mode from 46 tied sets.

The group of the top-ranked mode does not depend on the order: group accuracy is 0.342 under every tie-break rule, because no tie crosses a group. As a set, the group contains the true mode in 34.2% of cycles with 2.34 modes on average, while the same number of modes taken from the top of the prevalence list contains it in 38.5%. The group is ahead only on coverage averaged over categories (0.243 against 0.222). The knowledge base fixes the size of the answer set, and the group of the top-ranked mode is less often correct than a choice by frequency alone.

6. TEST SELECTION OVER REPRESENTATION DIMENSIONS

For each non-isolable group we search for a repair: statistics of the existing signals with which a classifier, the *resolver*, separates the members.

Dimensions. The classifiers in this section read statistics of z. The retrieval uses only the peak magnitude $\max|z|$; *mag* is this peak on each of the 26 channels. The candidate dimensions are further statistics of z: *sign* (mean of z), *step* (last third of the cycle minus first third), *payload* (carrying minus empty actions), *shape* (standard deviation of z), and *unread* (peak, signed, and payload statistics of two recorded channel families outside the 26). Each has a cost class: *free* if computed from the 26 channels, *pipeline* if it needs other recorded channels, *hardware* if it needs a measurement the cell does not have. None of our dimensions is of hardware class.

Search and certificate. Every learned component is a standardised ℓ_2 logistic regression ($C = 1$). Starting from mag, a greedy search adds the dimension whose gain in grouped inner cross-validation accuracy on the training half most exceeds 0.005 times its cost factor (1 for free, 6 for pipeline), and stops when no gain does. The inner validation has $\min(4, n)$ folds, with n the fewest training settings of any member. The selected set is adopted on a split if its inner accuracy exceeds the share of the most frequent member in the training half by 0.05 and the accuracy of mag alone by 0.02; the repair is adopted for the group if this holds on a majority of splits. If a member has fewer than two instantiations in the training half, no search can run; we then evaluate the set of all free dimensions and do not adopt. The test half is used only for the certificate, which gives the dimensions, their cost class, the held-out accuracy within the group, and three controls (accuracies in Table 2). Varying the cost factor, the minimum gain, and both margins does not change for which group a repair is adopted (repository).

Adopted. Within the load-torque group the repair reaches 0.840 on held-out instantiations, against 0.562 for the deployed features, 0.504 for the majority class, and 0.431 with permuted labels, and it is adopted on all five splits. The dimension selected most often is sign (4 of 5 splits; step on 3, unread and shape on 2 each); mag with sign alone reaches 0.827. Because unread enters on two splits, the certificate's worst-case cost class is pipeline, but the free set {sign, step} reaches 0.869 against 0.872 for the best set with unread channels, so no additional channel is needed. With {sign, step} most remaining errors are between the two payload modes: on cycles of these two modes the resolver is right in 0.80 of cases, against 0.93 and 0.98 for the pairs with axis weight. These figures are computed on cycles whose true mode is in the group, so they state how well the repair works once a cycle reaches the group.

Not adopted. Carton collision has three instantiations, so each training half contains one and no grouped search can run. The set of all free dimensions reaches 0.776 against a majority rate of 0.718. Each split trains on one carton instantiation and tests on the other two, and three of the five splits test on the same two, so the splits are not independent replications. Sec. 7 compares the protocols on this group with folds on which the search can run.

Undetermined. Each gripping mode has one instantiation, so no grouped split can test a repair.

Untestable. voraus-AD contains no over-temperature fault, so no search is run. The recommended check for over-temperature names a separator: current against torque for the same motion.

End to end, the two-stage system (retrieval, then the resolver for cycles whose top group is load torque) raises top-1 in declared order from 0.169 ± 0.018 to 0.302 ± 0.047 with dimensions searched on each split. Fixing the dimensions selected on most splits, {sign, step}, gives 0.309 ± 0.031, balanced recall 0.208 (from 0.105), and 0.869 within the group. That choice used all five splits; made from the other four splits each time, it gives 0.301 and 0.846, lower by 0.008 and 0.023. The gain of the fixed configuration over the retrieval is positive on every split, from 0.126 to 0.154. It comes from the resolver, not from the routing: sending every cycle to the load-torque resolver gives 0.349 and balanced recall 0.237, more than the two-stage system on all five splits.

On the test halves the load-torque group is the true answer for 40.1% of cycles but is ranked first for 66.1%. The retrieval routes 69% of the load-torque cycles to the group, and 58% of the cycles it routes there are other faults. Of the routed cycles the resolver labels 0.37 correctly (0.88 of the load-torque cycles among them), against 0.16 for the retrieval alone. Sending every cycle to the resolver also reaches the 31% of load-torque cycles that the retrieval ranks elsewhere. No cycle reaches the collision or gripping group (Sec. 4), so their certificates hold only for cycles that would reach them.

7. GROUPED AND RANDOM VALIDATION

Each `setting` value is one physical fault instantiation. A random split over cycles therefore places cycles of the same instantiation in both halves, and a model can learn to recognise the instantiation instead of the fault. The effect depends on whether a method learns from labels. The flat classifier of Sec. 8 falls from 0.800 under a random split to 0.525 under leave-setting-out. The retrieval, which learns nothing from the training half, gives 0.161 and 0.162 over 20 splits of each kind.

For the load-torque group the repair is adopted under both protocols (Table 2). For the collision group the protocols disagree. Under a random split the inner validation, whose folds share instantiations, reaches 0.85 to 0.96, and the repair is adopted on all five splits, although on held-out cycles it exceeds the majority rate of the same split by only 3 of 47 cycles (0.830 against 0.766). Under the 50/50 leave-setting-out split no grouped search can run (Sec. 6), so we compare the protocols with nine folds and the same search and rule. With folds stratified over cycles, the repair is adopted on 9 of 9 folds. With one setting held out per fold (leave-one-setting-out), it is adopted on 4 of 9 folds, and its pooled held-out accuracy is 0.67 to 0.74 over ten inner seeds, below the majority rate of 0.766 for every seed.

Table 2. Repair certificates: accuracy within the group on held-out cycles of its members. Grouped: leave-setting-out, the protocol we report. Ungrouped: the same code under a random split over cycles, including its inner validation. Deployed: the same classifier on the mag features alone. Majority: accuracy of always naming the training-half majority class. Permuted: the selected dimensions refitted on training labels shuffled over cycles. Deployed, Majority, and Permuted are grouped. †One carton instantiation in each training half, so no grouped search can run; the value is for the set of all free dimensions.

Group	Ungrouped	Grouped	Deployed	Majority	Permuted	Verdict
load torque (3 modes)	0.933	0.840	0.562	0.504	0.431	**adopted** on 5/5 splits, both protocols
distal collision (2 modes)	0.830	0.776†	0.709	0.718	0.608	**not adopted**: search cannot run; adopted ungrouped
gripping (2 modes)	one instantiation per member					undetermined
commutation/temp (2 modes)	no over-temperature instance in voraus-AD					untestable

A permuted-label control that shuffles labels over cycles scores below the collision repair on all five splits (0.53 to 0.81 against 0.83), so this control passes the repair. Shuffled over cycles, the labels are no longer tied to the instantiation, so the permuted model cannot gain from recognising instantiations. Permuting labels over whole settings keeps each label tied to an instantiation. Under the random split, 5 to 10 of the 84 possible labellings then reach the observed collision accuracy on each split ($p = 0.06$ to 0.12), so this control does not support the collision repair, while none of 300 labellings reaches the load-torque accuracy ($p < 0.004$). Any learned comparison on this benchmark should therefore group by setting in the inner validation as well as in the outer split.

8. CONTROLS AND LIMITS

8.1 The flat classifier

A flat 12-way logistic regression over all dimensions reaches 0.525 top-1, 0.397 balanced recall, and 0.574 group accuracy under the same protocol, against 0.309, 0.208, and 0.342 for the two-stage system. It recognises foam collision in 0.506 of cases, where the retrieval and the prior both score zero, so the features contain evidence that the retrieval does not use. We tested whether the knowledge base reduces the number of labels needed. Over 20 paired leave-setting-out splits, the classifier is trained on k labelled cycles and the two-stage system on k labelled cycles per group; only the load-torque resolver receives test cycles, so only its labels are used. Both are scored on the 12-way task. At $k = 3$ the two-stage system, which is then close to the knowledge base alone, has the higher balanced recall on 14 of 20 splits. From $k = 8$ the classifier has the higher top-1 (0.225 against 0.208, 15 of 20 splits) and balanced recall (0.154 against 0.136, 15 of 20), and at $k = 320$ it reaches 0.550 against 0.304. If the two-stage system receives the same labelled cycles as the classifier, it is behind at $k = 8$ on all 20 splits. With 8 or more labels the flat classifier is the better choice.

8.2 What would settle the open verdicts

Gripping needs at least two instantiations per mode for any grouped protocol, and voraus-AD has one. For collision the limit is our split: a 50/50 grouped split leaves $\lfloor N/2 \rfloor$ instantiations per mode for training, the grouped inner search needs two, and carton has $N = 3$. Under leave-one-setting-out the search can run, and the repair is still not adopted (Sec. 7). Sweeps over N on injected and simulated faults (repository) do not show how many instantiations a real pair needs, because there the deployed features already separate the pair.

9. CONCLUSION

On voraus-AD the ranked list of an FMEA-based diagnosis layer is limited by the structure of its knowledge base: 68.1% of anomalies tie at rank one, almost all in the load-torque group. A single top-1 figure then depends on the tie-break order more than on the split, and a prevalence prior is better on top-3, top-5, MRR, and group accuracy under every order.

We therefore report the non-isolable group and search for a statistic that separates it, with held-out evidence, controls, and a cost class. One of the three groups with benchmark instances is separated on unseen instantiations, mainly by the sign of the deviation, at no extra cost. This search must be validated by instantiation: under a random split it adopts a collision repair that leave-one-setting-out validation does not, and a permuted-label control over cycles passes it.

Isolability can be checked before any data are seen, but removing ties does not make the list more accurate: scoring the declared direction and time course removes them and lowers accuracy. With 8 or more labelled cycles a flat classifier over the same features is more accurate than the two-stage system. A change of representation should be adopted on evidence from unseen instantiations, not because it removes ties.

Code and data. Evidence records, certificates, all analyses and the producing scripts are archived at `https://doi.org/10.5281/zenodo.22089173`.

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